Lab 4: VibeCoding — Lab Reflection

In this lab, I worked on an Endless Runner game using AI-assisted coding. I started by inspecting the existing code, understanding the issues, and fixing them one task at a time using prompts. I improved collision detection and limited the speed increase to prevent obstacles from passing through the player without registering a collision. I then implemented a Game Over screen displaying the final score, added replay functionality with Easy, Medium, and Hard difficulty levels, and integrated sound effects for jumping, scoring, and game over.

This lab helped me understand how to use AI coding tools effectively by providing clear, specific prompts and making incremental changes instead of trying to implement everything at once. I also learned the importance of reviewing generated code, testing each feature, debugging errors, and maintaining separate Git commits for individual tasks. Overall, this lab improved my understanding of Python, Pygame, debugging, version control, and the practical use of AI-assisted software development.